

KHMELNYSKYI - SCENARIO BUILDING WORKSHOP

Objective: To collaboratively identify the key factors and drivers shaping the future of Khmelnytskyi's AI-powered transport management pilot and co-create plausible future scenarios for scaling digital and climate-smart governance, grounded in local uncertainties around technology deployment and long-term financing.

Uncertainty 1: AI Infrastructure Affordability: Affordable AI systems and infrastructure deployment vs. High implementation and infrastructure costs. Why? The workshop identified the high cost of AI systems, computing infrastructure, and local data storage as a major barrier to implementing and scaling AI-enabled municipal service

Uncertainty 2: Long-term Funding Security – Stable long-term funding and investment vs. Dependence on short-term external funding. Why? The workshop highlighted that scaling the pilot depends on securing sustainable funding beyond externally funded projects and grants.

Part I

Scenario Matrix for Khmelnytskyi's Adaptive traffic management Pilot

Long-term Funding Security					
Scenario C Low Affordability + Strong Funding Security			Scenario A High Affordability + Strong Funding Security		
<i>Future conditions and implications</i>	<i>Implementation and institutionalisation</i>	<i>Preparation and immediate actions</i>	<i>Future conditions and implications</i>	<i>Implementation and institutionalisation</i>	<i>Preparation and immediate actions</i>
What does this future look like for the pilot? How is pilot functioning in this future - Is it expanding, stagnating, adapting, transforming - What opportunities and challenges emerge - Key actors and most vulnerable groups	What would be needed to make this successful? What policies are required - What technical capacities are required - What financing mechanisms are required - What governance arrangements are required	What should the city start doing today? What actions should begin immediately - What capacities need strengthening - What partnerships should develop - What governance structures should be established?	What does this future look like for the pilot? How is pilot functioning in this future - Is it expanding, stagnating, adapting, transforming - What opportunities and challenges emerge - Key actors and most vulnerable groups	What would be needed to make this successful? What policies are required - What technical capacities are required - What financing mechanisms are required - What governance arrangements are required	What should the city start doing today? What actions should begin immediately - What capacities need strengthening - What partnerships should develop - What governance structures should be established?
Scenario D Low Affordability + Weak Funding Security			Scenario B High Affordability + Weak Funding Security		
<i>Future conditions and implications</i>	<i>Implementation and institutionalisation</i>	<i>Preparation and immediate actions</i>	<i>Future conditions and implications</i>	<i>Implementation and institutionalisation</i>	<i>Preparation and immediate actions</i>
What does this future look like for the pilot? How is pilot functioning in this future - Is it expanding, stagnating, adapting, transforming - What opportunities and challenges emerge - Key actors and most vulnerable groups	What would be needed to make this successful? What policies are required - What technical capacities are required - What financing mechanisms are required - What governance arrangements are required	What should the city start doing today? What actions should begin immediately - What capacities need strengthening - What partnerships should develop - What governance structures should be established?	What does this future look like for the pilot? How is pilot functioning in this future - Is it expanding, stagnating, adapting, transforming - What opportunities and challenges emerge - Key actors and most vulnerable groups	What would be needed to make this successful? What policies are required - What technical capacities are required - What financing mechanisms are required - What governance arrangements are required	What should the city start doing today? What actions should begin immediately - What capacities need strengthening - What partnerships should develop - What governance structures should be established?

Part II

Cross-Scenario Reflection

Which readiness gaps appear repeatedly in scenarios?
Which URI dimensions remain weak regardless of future? What barriers repeatedly limit success?

1. Vision & Strategic
vision / political commitment / EU alignment / Environmental policy / pilot readiness

3. Institutional & Technical
institutions / risk assessment / project delivery / digital systems / research partnerships

5. Innovation & Infrastructure
infrastructure base / renewables / efficiency/retrofit programs / circular systems / innovation ecosystems

2. Regulatory & Planning
climate legislation / master planning / zoning / transport planning / waste systems

4. Financial & Economic
municipal finance / EU funding track record / diverse green economy / industry transition / PPPs for climate

6. Civic & Stakeholder
citizen engagement / public awareness / participation tools / education/skills / NGOs

7. Conflict Impact & Recovery
conflict exposure / damage assessment / recovery planning / adaptation strategy / heritage protection